Explicit polynomial solutions of Cohen's two difference equations

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Abstract

We prove the existence assertion in Cohen's Conjecture 4.2 for every nonnegative integer parameter. An explicit finite polynomial is constructed by a residue of a terminating hypergeometric expression. Together with the previously observed degree, uniqueness, and reflection properties, this settles both equations in MathDB 358171. The proof uses two symbolic coefficient identities and does not depend on a finite computational search.

Task 1: the conjecture and its complete proof

For $u \in \mathbb{Z}_{\geq 0}$, put

$$N = u(u+3), \qquad n = N/2, \qquad \lambda = 2u^2 + 8u + 7, \qquad \mu = (2u+3)(N+1).$$

The number N is even. Cohen's Conjecture 4.2 [1], on pages 7–8, asks for nonzero polynomial solutions of

$$\begin{aligned}(x+2u+4)(x+u+1)P(x+1) - (x-1)(x+u-1)P(x-1) \\ = (\lambda x + \mu)P(x),\end{aligned} \tag{1}$$

$$\begin{aligned}(x+2u+4)(x+u+4)Q(x+1) - (x-1)(x+u+2)Q(x-1) \\ = \big(\lambda x + (2u+3)(u+2)(u+3)\big)Q(x).\end{aligned} \tag{2}$$

The conjecture predicts degree N and uniqueness up to nonzero scalar multiples for each equation. The degree, uniqueness conditional on existence, and equivalence of the two equations are already due to Peter Mueller [2]; they are also credited on page 8 of [1]. We supply the missing existence proof and repeat those short arguments to make the result self-contained.

Theorem 1. *For every nonnegative integer u, the solution space in $\mathbb{Q}[x]$ of each equation* (1) *and* (2) *is one-dimensional, and every nonzero solution has degree $N = u(u+3)$.*

The explicit construction

Define

$$D = N + 2u + 5, \qquad K = N + u + 3 = (u+1)(u+3), \qquad e_k = \frac{(-n)_k (1/2)_k}{(-D/2)_k \, k!} \quad (0 \leq k \leq n)$$

and

$$H(z) = \sum_{k=0}^{n} e_k \left(z^{2k} + \frac{2k-K}{2k-D} z^{2k+1} \right) = \sum_{j=0}^{N+1} h_j z^j. \tag{3}$$

All coefficients are rational and all denominators are nonzero: $D > N$, and $-D/2$ is a negative half-integer. The required polynomial is

$$P(x) = \operatorname*{Res}_{t=0} e^{(x+u)t} H\big(\coth(t/2)\big)\, dt. \tag{4}$$

This means the coefficient of t^{-1} in the indicated Laurent series. Equivalently, it is the following completely finite expression:

$$P(x) = \sum_{j=1}^{N+1} h_j \sum_{\ell=0}^{j-1} 2^{\ell+1} \binom{j}{\ell+1} \binom{x+u-1}{\ell}, \qquad Q(x) = P(-x-2u-3). \tag{5}$$

Here the generalized binomial is a polynomial in its upper argument. The equivalence follows from $w = e^t$, $v = w - 1$:

$$P(x) = \operatorname*{Res}_{v=0} (1+v)^{x+u-1} H\big((v+2)/v\big)\, dv.$$

Expanding each finite power $(v+2)^j$ gives (5).

A polynomial differential equation

Lemma 1. *The polynomial* (3) *satisfies*

$$z(z^2-1)H''(z) + (-Nz^2 + 3z + D)H'(z) - KH(z) = 0. \tag{6}$$

Proof. For any coefficients h_j, the coefficient of z^j on the left of (6) is

$$(j-1)(j-2-N)h_{j-1} + (3j-K)h_j + (j+1)(D-j)h_{j+1}. \tag{7}$$

Take coefficients outside $0 \le j \le N+1$ to be zero. Write $o_k = h_{2k+1}$, so that

$$h_{2k} = e_k, \qquad o_k = \frac{2k-K}{2k-D} e_k, \qquad \frac{e_{k+1}}{e_k} = \frac{(2k-N)(2k+1)}{2(k+1)(2k-D)}.$$

The last relation also gives $e_{n+1} = 0$. For $j = 2k$, substitution in (7) reduces to

$$\big[2k(2k-2-K) + (6k-K) - (2k+1)(2k-K)\big] e_k = 0.$$

At $k = 0$ the first term is absent and the same identity applies. For $j = 2k+1$, multiply (7) by $(2k-D)/e_k$. The result is

$$\begin{aligned} 2k(2k-1-N)(2k-D) + (6k+3-K)(2k-K) \\ + (D-2k-1)(2k-N)(2k+1). \end{aligned}$$

Using $D = N + 2u + 5$ and $K = N + u + 3$, this expression is

$$u^2 + 3u - N = 0.$$

This includes $k = n$, where the last term vanishes. The only remaining possible coefficient is $j = N+2$; its factor $j - 2 - N$ is zero. Higher coefficients vanish. This proves the differential equation for every u. □

Transporting the differential equation to a residue

Set

$$g(w) = w^u H\left(\frac{w+1}{w-1}\right), \qquad f(t) = g(e^t) = e^{ut} H\big(\coth(t/2)\big).$$

The chain rule transforms (6) into

$$w(w^2-1)g'' + \big(-(3u+2)w^2 + \lambda w + u - 1\big)g' + \big(u(2u+3)w - \mu\big)g = 0. \tag{8}$$

Here are explicit coefficient checks for that transformation. With $z = (w+1)/(w-1)$, let $a = w(w^2-1)$, $b = -(3u+2)w^2 + \lambda w + u - 1$, and $c = u(2u+3)w - \mu$. After division by w^u, the coefficients of H'', H', H are

$$a(z')^2, \qquad a\left(z'' + \frac{2u}{w}z'\right) + bz', \qquad \frac{u(u-1)}{w^2}a + \frac{u}{w}b + c.$$

Substitution of $w = (z+1)/(z-1)$ gives, respectively, $z(z^2-1)$, $-Nz^2+3z+D$, and $-K$. In turn, $w = e^t$ converts (8) into

$$(e^t - e^{-t})f'' + \big(\lambda - 3(u+1)e^t + ue^{-t}\big)f' + \big(u(2u+3)e^t - \mu\big)f = 0. \tag{9}$$

We now verify (1) directly. Set

$$A(x) = x^2 + (3u+5)x + (2u+4)(u+1), \qquad B(x) = x^2 + (u-2)x + 1 - u.$$

For any meromorphic germ $F(t)$,

$$\operatorname*{Res}_{t=0}(e^{xt}F(t))'\,dt = 0, \qquad x\operatorname{Res} e^{xt}F\,dt = -\operatorname{Res} e^{xt}F'\,dt.$$

Applying this identity twice shows that the left side of (1) minus its right side equals the residue of

$$e^{xt}\left\{A(-\partial_t)(e^t f) - B(-\partial_t)(e^{-t}f) + \lambda f' - \mu f\right\}dt.$$

Expanding the braces gives exactly the left side of (9). That expression vanishes identically, proving (1). All operations concern germs with finite-order poles, so no limiting integral, summability, or exchange of infinite tails is needed.

Nonvanishing, degree, uniqueness, and the second equation

The leading coefficient of H is

$$h_{N+1} = e_n\frac{u+3}{2u+5} \neq 0.$$

Since $\coth(t/2) = 2/t + O(t)$, equation (4) is a polynomial of degree exactly N, with leading coefficient

$$[x^N]P(x) = \frac{2^{N+1}h_{N+1}}{N!} \neq 0.$$

Its coefficients are rational, also directly evident from (5). This proves the previously unresolved existence.

For clarity, the short uniqueness argument from [2] is as follows. If a nonzero polynomial of degree d, with leading coefficient a, satisfies (1), the coefficient of x^{d+1} in the residual is

$$a(2d + 2u + 7 - \lambda) = 2a(d - N).$$

Thus $d = N$. Two monic solutions would have a difference of smaller degree, hence must coincide. Finally, substitute $y = -x - 2u - 3$ into (1) and multiply by -1. The two shifted coefficients interchange and

$$\lambda(2u+3) - \mu = (2u+3)(u+2)(u+3).$$

Therefore $Q(x) = P(-x-2u-3)$ satisfies (2). This substitution is invertible, so its solution space is also one-dimensional. Theorem 1 follows.

For example, the monic solutions of (1) at $u = 0, 1$ are

$$1, \qquad x^4 + \frac{15}{2}x^3 + \frac{85}{2}x^2 + 105x + 144.$$

The construction also works for real $u \geq 0$ whenever $u(u+3)$ is an even integer; then the coefficients lie in $\mathbb{Q}(u)$. This follows from the same proof and earns no additional problem credit.

Task 2: remove the logarithmic term in Cohen's application

Using the explicit P, settle the separate assertion on page 8 of [1] that the associated alternating rational series has no $\log 2$ term. Verify the coprimality conditions needed for Proposition 4.1 and obtain the initial constants in the resulting continued fractions for π^2.

Task 3: determine the arithmetic of the monic solution

Determine the least integer clearing the denominators of the monic P, including its prime valuations as functions of u. Test and, if true, prove positivity of its ordinary coefficients and irreducibility over $\mathbb{Q}$. These properties are not assumed in the existence proof.

Task 4: classify the full parameter locus

For even $N \geq 0$, classify all pairs D, K for which

$$z(z^2-1)H'' + (-Nz^2 + 3z + D)H' - KH = 0$$

has a polynomial solution of degree $N+1$. The construction identifies the component $D = N+2u+5,\ K = N+u+3,\ N = u(u+3)$. Describe additional components, exceptional denominators, and their corresponding difference equations.

Task 5: use the hypergeometric form to study zeros

The even part of H is ${}_2F_1(-N/2, 1/2; -D/2; z^2)$. Use this terminating representation and its contiguous odd part to locate the zeros of H and P, and to derive their scaled zero distributions as u tends to infinity. Determine whether these formulas yield faster exact construction than solving for the coefficients by descending elimination.

References

[1] H. Cohen, *Parametric Continued Fractions for π^2, $\zeta(3)$, and other Constants*, arXiv:2304.11727v1 (2023), `https://arxiv.org/abs/2304.11727`. Conjecture 4.2 and the credited partial results are on pp. 7–8.

[2] P. Mueller, answer to H. Cohen, *Polynomial solutions to a difference equation*, MathOverflow, 10 April 2023, `https://mathoverflow.net/questions/444367/`.